MMSU Case Study Report Template

Designing an Institutional Information Management (IM) Framework and System Plan

Title Page

- **Content:** Include the title of your case study, course name, section, names of group members (4–5), instructor's name, date of submission, and institution.
- **Your** title should reflect both the organization studied and the focus of the report (e.g., *"Information Management Framework for the College of Information Technology at MMSU"*).

I. Introduction

Introduce the chosen organization and explain why IM is important. State purpose, scope, and expected outcomes.

Purpose of this section:
Set the stage for your study by introducing the organization, the purpose of your analysis, and the significance of managing information effectively.

Write about:

- A short description of your chosen organization or unit (e.g., college, department, small business).
- Why information management is important to this organization.
- The goal of your case study — for example, "to propose an institutional IM framework that enhances data governance, quality, and compliance."
- The scope and limitations of your study (what parts of the organization's information environment you examined).

II. Organizational Profile and Information Environment

Describe mission, functions, data types, current systems, and information flow issues.

Purpose of this section:
Provide context about the organization and its current information landscape.

Write about:

- **Background:** Organizational mission, services, and key functions.
- **Data Landscape:** What kinds of data the organization handles (e.g., student data, HR records, transactions).
- **Existing Systems:** List or describe current systems, applications, or databases in use.
- **Pain Points:** Identify issues such as duplication, lack of integration, outdated databases, or poor data sharing.

Present the data flow in paragraph form first, then summarize it in a diagram or simple table.

III. Information Architecture

Explain how information is organized, stored, and flows across systems. Include an Information Architecture Diagram.

Purpose of this section: Illustrate how information flows through systems, applications, and data repositories in the organization.

Write about:

- **Architecture Overview**: Explain how information is created, stored, processed, and accessed.
- **Systems Mapping**: Identify data-producing systems and how they interact (e.g., Registrar System → MDM Hub → Analytics Dashboard).
- **Diagram**: Include an architecture diagram showing data sources, central repositories, and users.
- **Commentary:** Discuss how this structure supports or limits organizational efficiency.

Use a visual (e.g., PowerPoint, Draw.io, Lucidchart) for clarity. Add labels like Data Source, Processing Layer, MDM Hub, Analytics Tools.

IV. Master Data Management (MDM) Framework

Identify master data, describe the chosen MDM model, and explain how data is standardized, synchronized, and governed.

Information Architecture (Part III) describes the STRUCTURE—how systems connect and how data flows.

The MDM Framework (Part IV) describes the CONTROLS—how master data is standardized, synchronized, and governed. Architecture is the MAP; MDM is the RULEBOOK.

Purpose of this section:
Show how you propose to organize and manage shared, critical data assets.

Write about:

- **Concept:** Define what master data means for your case organization (e.g., student records, employee data).
- **MDM Model:** Describe whether your framework is centralized, federated, or hybrid — and why it fits your organization's needs.
- **Components:** Identify key elements such as data integration tools, validation processes, and metadata management.
- **Workflow:** Explain how data will be cleaned, standardized, and synchronized across systems.
- **Illustration:** Include a conceptual diagram showing how data flows between the MDM hub and other systems.

Emphasize how MDM reduces redundancy and enhances data reliability.

V. Data Governance and Quality Policy

Define governance roles, rules, and data quality standards. Submit the completed RACI Template as Appendix A.

Purpose of this section:
Outline how your organization should establish accountability and standards for managing information.

Write about:

- **Governance Structure:** Identify data-related roles — e.g., Data Owner, Data Steward, Custodian, End User.
- **Policies:** Describe policies on data access, sharing, retention, and documentation.
- **Quality Dimensions:** Explain how accuracy, completeness, consistency, and timeliness will be maintained.
- **Implementation Plan:** Provide short recommendations for governance adoption and monitoring.

VI. Privacy, Regulatory, and Compliance Considerations

Explain RA 10173 compliance and describe privacy risks and controls.

Purpose of this section:
Demonstrate awareness of legal and ethical responsibilities in managing data.

Write about:

- **Applicable Laws:** Reference RA 10173 (Data Privacy Act of 2012) guidelines.
- **Privacy Measures:** Describe how your framework ensures confidentiality, data minimization, consent, and breach notification.
- **Compliance Controls:** Suggest actions such as data classification, privacy notices, or encryption.
- **Ethical Considerations:** Discuss responsible data handling in academic or institutional contexts.

Use institutional privacy manuals as references, if any.

VII. Information Security and Risk Management

Identify risks, rate likelihood and impact, and propose controls. Submit the completed Risk Matrix as Appendix B.

Purpose of this section:
Identify potential security risks and propose mitigation strategies.

Write about:

- **Threats and Vulnerabilities:** Examples — unauthorized access, malware, insider threats, data leaks.
- **Risk Assessment Summary:** Briefly rank risks by likelihood and impact.
- **Controls:** Recommend administrative (policies), technical (firewalls, encryption), and physical (secure storage) controls.
- **Continuity Plan:** Suggest backup, disaster recovery, and incident response actions.

VIII. Integration and Use for Decision-Making

Explain how integration improves collaboration and service delivery. Submit the completed Integration Matrix as Appendix C.

Purpose of this section: Show how the IM framework supports analytics, reporting, and strategic goals.

Write about:

- How the IM system enables evidence-based decision-making.
- Examples of reports or dashboards that the organization could generate.
- How data integration improves collaboration and service delivery.
- Describe one scenario where better data management leads to tangible improvement (e.g., faster reporting, reduced errors).

Connect this to organizational impact, not just technical benefits.

IX. Summary of Findings and Recommendations

Provide consolidated insights and actions based on your analysis.

Purpose of this section:
Wrap up your study by summarizing what you learned and the improvements your IM framework provides.

Write about:

- Key insights from your analysis.
- The most important improvements or efficiencies your proposal introduces.
- Any limitations encountered during your study (e.g., limited data access).
- Future actions or tools your organization could adopt to sustain IM maturity.

Keep it concise — focus on value-added results.

X. References

List all sources in APA 7th edition.

Cite all books, frameworks, and online sources used. Add RA 10173/RA 10173 documents as supporting sources.

XI. Appendices

Attach diagrams, RACI, Risk Matrix, Integration Matrix, and supporting files.

Include supplementary materials such as:

- Data flow diagrams
- Sample policy templates
- Governance matrices
- Interview notes or survey instruments (if any)

FORMATTING REQUIREMENTS

- Length: 10–15 pages, A4, 1.15 spacing, Calibri 11.
- Headings: Follow the numbering in this template.
- Visuals: Label all figures and tables (e.g., “Figure 1. Information Architecture of the College”).
- Submission: PDF, printed, with title page and group details.

Template A — RACI Matrix (Data Governance)

Purpose: To assign roles using R (Responsible), A (Accountable), C (Consulted), and I (Informed). Each activity must have exactly one 'A'. Fill based on your organization's structure.

Activity	Data Owner	Data Steward	Data Custodian	Data User	DPO	Administrator
Data Classification						
Data Quality Monitoring						
Metadata Management						
Access Rights Assignment						
Data Update & Correction						
Data Sharing Requests						
Privacy Compliance Checks						
Incident Response						
Backup & Recovery						
Policy Review & Updates						

Template B — Risk Matrix (Information Security & Risk Management)

Purpose: To identify risks, score likelihood and impact, determine the risk level, and recommend controls. Fill all fields accurately.

Risk/Threat	Asset	Vulnerability	Likelihood (1–5)	Impact (1–5)	Risk Score	Risk Level	Controls

Risk Matrix Grid

	1	2	3	4	5
5 – Almost Certain	M	M	H	H	H
4 – Likely	M	M	M	H	H
3 – Possible	L	M	M	M	H
2 – Unlikely	L	L	M	M	M
1 – Rare	L	L	L	L	M

Template C — Integration Matrix (Decision-Making & Service Delivery)

Purpose: To map workflow steps, identify units and systems involved, and show how integration improves service delivery and supports better decision-making. Fill each row clearly.

Workflow Step	Unit / Office	System / Data Source	Data Produced / Shared	Impact of Integration

1. **SAMPLE RACI MATRIX (Data Governance)**

Scenario: "Student Academic Records" as master data

Data Governance Activity	Data Owner (Registrar)	Data Steward (Records Supervisor)	Data Custodian (IT Unit)	Data User (Dept. Chairs, Faculty)	DPO / Compliance	Administrator (VPAA)
Data Classification	A	R	C	I	C	I
Data Quality Monitoring	A	R	C	C	I	I
Metadata Management	C	R	C	I	I	A
Access Rights Assignment	A	C	R	I	C	I
Data Update & Correction	A	R	C	R (initiate correction requests)	I	I
Data Sharing Requests	A	C	R	I	C	I
Privacy Compliance Checks	I	C	C	I	A/R	I
Incident Response	A	C	R	R (first reporter)	C	I
Backup & Recovery	I	I	R/A	I	C	I
Policy Review & Updates	A	R	C	C	C	A (final approval)

2. SAMPLE RISK MATRIX (Information Security & Risk Management)

Scenario: Risks in handling student data for the IM framework

A. Risk Identification & Scoring

Risk / Threat	Asset Affected	Vulnerability	Likelihood (1–5)	Impact (1–5)	Score	Risk Level	Controls / Treatments
Unauthorized access to student grades	Student Information System	Weak password practices	4	4	16	High	MFA, password policy, account review
Data loss due to hardware failure	Academic Records Database	No automated backup	3	5	15	High	Daily backups, offsite backup, redundancy
Accidental data modification	Class Records	Insufficient user training	3	3	9	Medium	Access limits, versioning, training
Malware attack	Faculty workstations	Outdated antivirus	2	4	8	Medium	Up-to-date endpoint protection, email filtering
Delayed incident reporting	All student data	No reporting SOP	2	3	6	Medium	Create SOP, escalation chain, awareness

B. Risk Matrix Grid (for interpretation)

Likelihood ↓ / Impact →	1	2	3	4	5
5 – Almost Certain	M	M	H	H	H
4 – Likely	M	M	M	H	H
3 – Possible	L	M	M	M	H
2 – Unlikely	L	L	M	M	M
1 – Rare	L	L	L	L	M

3. SAMPLE INTEGRATION MATRIX (Integration & Decision-Making)

Scenario: Enrollment → Registrar → Accounting → MIS Dashboard

Workflow Step	Unit / Office	System / Data Source	Data Produced / Shared	Impact of Integration on Efficiency & Service Delivery
Student submits enrollment form	College Department	Pre-Enrollment Form	Initial student data (name, program)	Removes manual encoding; reduces errors
Department evaluates subjects	Department / Faculty	Curriculum Database	Subject approvals	Automatic subject validation prevents mismatches
Registrar validates and encodes	Registrar	Student Information System (SIS)	Official enrollment record	Real-time updates accessible across offices
Accounting computes fees	Accounting Office	Billing System + SIS	Billing statement	Immediate fee computation without re-encoding
Payment posted	Cashier	Finance System	Payment confirmation	Faster clearance; no repeated validation
Dashboard generates report	MIS/Planning	Integrated MDM Hub	Enrollment summary, trends	VPAA receives up-to-date analytics for decision-making